INTERSECTION NUMBER OF TWO ORTHOGONAL ALMOST COMPLEX STRUCTURES ON A CLOSED RIEMANNIAN SIX-MANIFOLD IN TERMS OF THEIR CHERN CLASSES

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Editorial note. Context blocks are the source results that the statement and its proof use: the cohomology presentation of the twistor spaces (label cohomtwis), the Poincare dual of the canonical section (label poincaredual), the comparison of the twistor class for two spin-c structures with its proof, and the labelled equation div2. They are marked as context and are not counted as this task's proof.

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Allowed standard prerequisites (editorial).

- Projective bundle formula. For a complex vector bundle E over M of rank k, H^*(P(E)) is the free H^*(M)-algebra on x = c_1(O_{E^*}(1)) subject only to x^k + c_1(E)x^{k-1} + ... + c_k(E) = 0.. the authors' own statement of the convention at source lines 168-169. It supplies the two presentations of H^*(Z_+(M)) in the variables x and x' that the retained computation rewrites and multiplies; no projective-bundle input beyond this is assumed.
- Poincare dual of the canonical section as a Thom class. For the canonical section s of P(E + C) given pointwise by [0:...:0:1], the Poincare dual of s_*[M] is x^n + c_1x^{n-1} + ... + c_n, the image of the Thom class of the normal bundle.. the article's own Lemma poincaredual, source lines 839-846, delivered here as a context span. It converts the two twistor sections into explicit degree-six classes in H^*(Z_+(M)); the retained obligation begins after this identification.

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- Spin-c structures as an H^2(M)-torsor. Isomorphism classes of spin-c structures on M form a torsor over H^2(M), and if [s'] = a . [s] then the Hopf bundles of the two projective-bundle models satisfy H' = H (x) L with c_1(L) = a, whence x' = x - a.. the article's own lemma and discussion at source lines 934-940 and 942-972, delivered here as a context span. It is exactly what licenses the substitution x' = x + (c_1 - c_1')/2 used in the proof; nothing else about spin-c structures is assumed.

Corollary 5.7. *Let M be an oriented Riemannian manifold with an orthogonal almost complex structure J compatible with the orientation. We have the following isomorphisms as algebras over $H^*(M)$:*

1. *If* $\dim(M) = 4$, *then*
$$H^*(Z_+(M)) \cong H^*(M)[x]/\big(x^2 + c_1 x\big),$$
$$H^*(Z_-(M)) \cong H^*(M)[x]/\big(x^2 + c_1 x + c_2\big).$$
2. *If* $\dim(M) = 6$, *then*
$$H^*(Z_+(M)) \cong H^*(M)[x]/\big(x^4 + 2c_1x^3 + \big(c_1^2 + c_2\big)x^2 + (c_1c_2 - c_3)x\big),$$
$$H^*(Z_-(M)) \cong H^*(M)[x]/\big(x^4 + 2c_1x^3 + \big(c_1^2 + c_2\big)x^2 + (c_1c_2 + c_3)x\big).$$

Lemma 5.8. *Let $E \xrightarrow{\pi} M$ be a complex vector bundle of rank n over a closed oriented manifold M. Let $s\colon M \to \mathbb{P}(E \oplus \mathbb{C})$ denote the canonical section given pointwise by $[0 : \cdots : 0 : 1]$. Then the Poincaré dual of $s_*[M]$ in $H^*(\mathbb{P}(E \oplus \mathbb{C})) \cong H^*(M)[x]/\big(x^{n+1} + c_1x^n + \cdots + c_nx\big)$ is*
$$x^n + c_1x^{n-1} + \cdots + c_n,$$
where c_i denote the Chern classes of E.[1]

Lemma 5.9. *Let M be an oriented Riemannian six-manifold and let $s, s'\colon M \to Z_+(M)$ be two almost complex structures on M compatible with the orientation of M. Let $a \in H^2(M)$ be (the unique cohomology class) such that $[s'] = a \cdot [s]$. Then the canonical isomorphism between the expressions for $H^*(Z_+(M))$ given in Corollary 5.7 in terms of s, s' is given by*
$$x' \mapsto x - a.$$

Proof. Let E and E' be the positive spinor bundles associated to the spin^c structures $[s]$ and $[s']$, respectively, and let L denote a complex line bundle with $c_1(L) = a$.

The relation $a \cdot [s] = [s']$ implies that there is an isomorphism
$$\psi\colon\ L \otimes E \to E'.$$
The kernel in $T_xM \otimes \mathbb{C}$ of right multiplication by a positive spinor $v \in E_x$ is the same as the kernel of right multiplication by $\lambda \otimes v \in L_x \otimes E_x$ for any $\lambda \neq 0$, so the canonical isomorphism given by the composition
$$\mathbb{P}(L \otimes E) \xrightarrow{\phi'} Z_+(M) \xrightarrow{\phi^{-1}} \mathbb{P}(E)$$
is the obvious one sending the line $L \otimes (\mathbb{C}v)$ to $\mathbb{C}v$.

We conclude that on $Z_+(M)$ the tautological (Hopf) line bundles H and H' coming respectively from the identifications with $\mathbb{P}(E)$ and $\mathbb{P}(E')$ satisfy the relation
$$H' \cong H \otimes L.$$
Since $x' = c_1(H'^*)$ and $x = c_1(H^*)$, this completes the proof. □

[1] Here the orientation on $\mathbb{P}(E \oplus \mathbb{C})$ is understood to be the one induced by the orientation on M and the canonical orientation on the fiber $\mathbb{CP}^n$ corresponding to $\langle x^n, [\mathbb{CP}^n]\rangle = 1$.

Given a spin^c structure $[s]$ we will write $c_1([s])$ for the Chern class of the line bundle classified by the projection $\mathcal{B}\mathrm{Spin}^c(2n) \xrightarrow{\mathcal{B}\delta} \mathcal{B}S^1$. Note that $c_1(a\cdot[s]) = 2a + c_1([s])$ so the difference class $a \in H^2(M)$ between two almost complex structures s, s' is a specific "square root" of the difference $c_1' - c_1$. We can suggestively write

$$x' = x + \frac{c_1 - c_1'}{2}, \tag{5.1}$$

which is unambiguous on rational cohomology (or when there is no 2-torsion in $H^2(M)$).

Theorem 5.10. *Let J and J' be two orthogonal almost complex structures inducing the same orientation on a closed Riemannian six-manifold M, corresponding to sections s and s' of the twistor bundle. Denote by c_i, c_i' their respective Chern classes. Then the intersection number $s_*[M]\cdot s'_*[M]$ in $Z_+(M)$ is given by*

$$\int_M \tfrac{1}{8}\big(c_1^3 + c_1^2c_1' - c_1c_1'^2 - c_1'^3\big) + \tfrac{1}{2}\big(c_1c_2' + c_1'c_2'\big) - c_3.$$

Proof. By Corollary 5.7, the presentation of $H^*(Z_+(M))$ with reference to J is given by

$$H^*(M)[x]/\big(x^4 + 2c_1x^3 + \big(c_1^2 + c_2\big)x^2 + \big(c_1c_2 - c_3\big)x\big),$$

and with reference to J' it is given by

$$H^*(M)[x']/\big((x')^4 + 2c_1'(x')^3 + ((c_1')^2 + c_2')(x')^2 + (c_1'c_2' - c_3')x'\big).$$

Combining Corollary 5.7 with Lemma 5.8, we have that the Poincaré dual of $s_*[M]$ is given by

$$x^3 + 2c_1x^2 + \big(c_1^2 + c_2\big)x + (c_1c_2 - c_3),$$

and the Poincaré dual of $s'_*[M]$ is given by

$$(x')^3 + 2c_1'(x')^2 + \big((c_1')^2 + c_2'\big)x' + c_1'c_2' - c_3'.$$

Recall from (5.1) that $x' = x + \frac{1}{2}(c_1 - c_1')$. (Our calculation will be insensitive to torsion, so we may in fact work in the rational cohomology ring where there is no ambiguity in writing $\frac{1}{2}(c_1 - c_1')$.) Using this to rewrite the expression for $s'_*[M]$, and multiplying with $s_*[M]$, yields

$$x^3\big(\tfrac{1}{8}\big(c_1^3 + c_1^2c_1' - c_1c_1'^2 - c_1'^3\big) + \tfrac{1}{2}(c_1c_2' + c_1'c_2') - c_3'\big).$$

Pairing with the fundamental class of $Z_+(M)$ gives the result. □